Computer Vision

Lecture 5

KeyPoints

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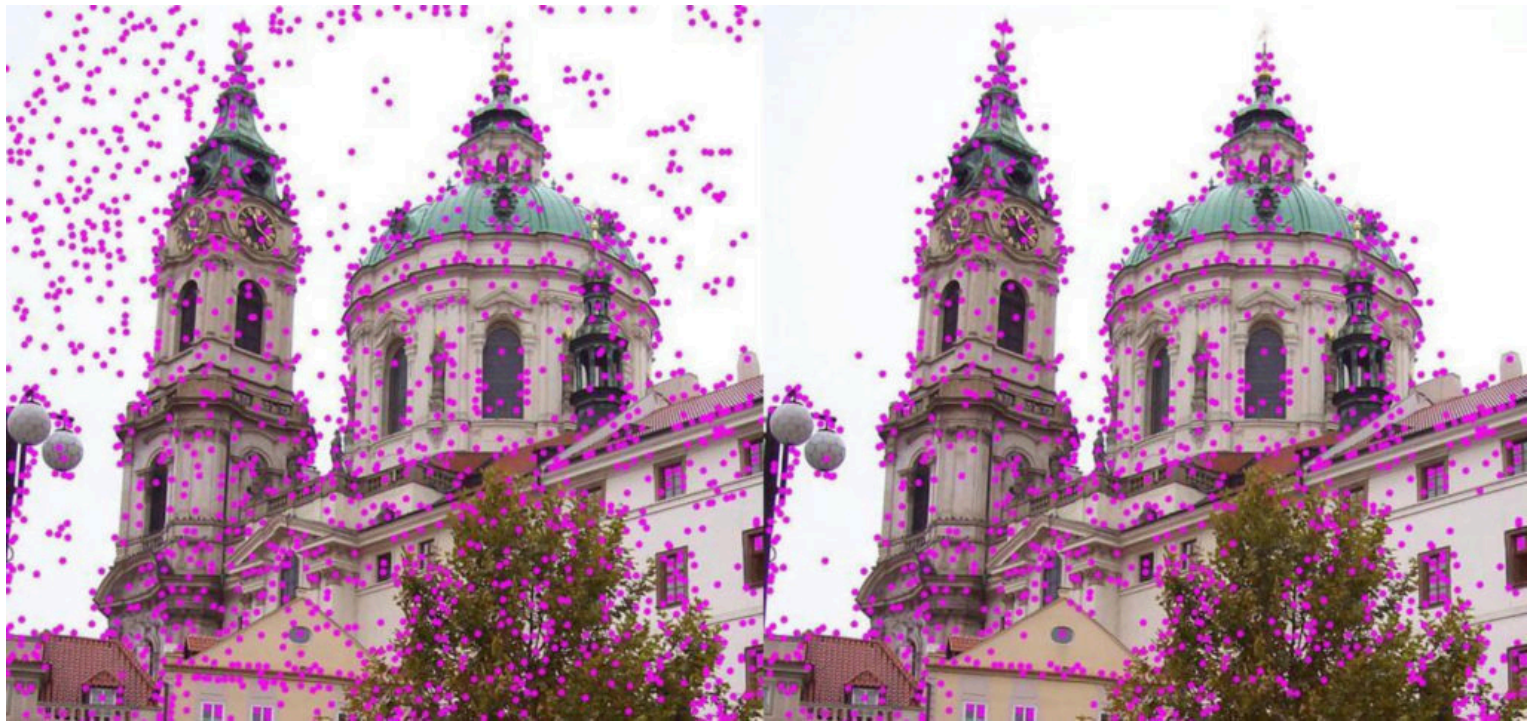
Introduction

What are keypoints?

- Not all points are equal. Some points, due to their position and values, contain rich information about the image.
- These points often appear near edges.
- Keypoint detection performs:
  - Locating important points in the image
  - Describing each point and its neighborhood using an easy-to-compare descriptor
  - Using these descriptors to compare points across two images to measure similarity

What are important points?

- A set of keypoints (example illustration)



Applications

- Content-based image retrieval:
  - Images are indexed by building a database of each image and its keypoints.
  - When searching for an image, keypoints are extracted from the query image and compared to the database.
  - The matching image is identified.
  - It is also possible to search for an image using only a part of it.
- Keypoints can help classify images (Bag of Visual Words algorithm).
- They are also used for object detection.
- Tracking
  - Used extensively in video.
  - Allows tracking a specific point across consecutive frames.
  - Useful for visual effects.

Keypoint Technologies

- Among the most popular keypoint descriptors:
  - SIFT
  - SURF
  - KAZE
  - ORB
- There are other ways to describe keypoints, however SIFT and SURF are most used

SIFT

- Scale-Invariant Feature Transform
- One of the earliest techniques for describing image information using keypoints.
- Developed in 2004.
- Highly accurate and robust to rotation, scale changes, and sometimes perspective.
- However, it is slow in extraction.
- Proprietary technology, when used outside of research, you’re required to pay royalties

Detection Stages

- SIFT detection consists of 4 stages:

  1. Scale-space extrema detection
  2. Keypoint localization
  3. Orientation assignment
  4. Keypoint descriptor

- A Difference of Gaussian (DoG) function is used to find stable keypoints in scale and orientation.

$$DoG(x, y, \sigma) = G(x, y, k\sigma) * I(x, y) - G(x, y, \sigma) * I(x, y)$$

Scale-space extrema detection

- A key aspect of SIFT is that it produces a large number of features densely covering the image across many scales and positions.
- To find scale-invariant keypoints, stacked Gaussian filters are used.
- Stable points across scales are selected.
- Scale-space is built using continuous Gaussian blurring.
- Given an image (I) and Gaussian kernel (G), convolution produces:

$$L(x, y, \sigma) = G(x, y, \sigma) * I(x, y)$$

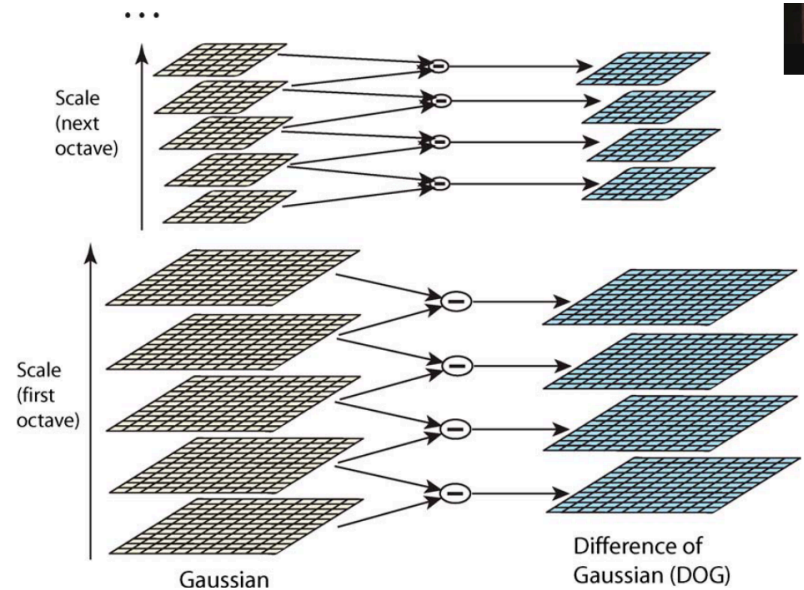
- The DoG filter approximates the Laplacian of Gaussian (LoG):

$$DoG(x, y, \sigma) = L(x, y, k\sigma) - L(x, y, \sigma)$$

- This is computationally cheaper than LoG.

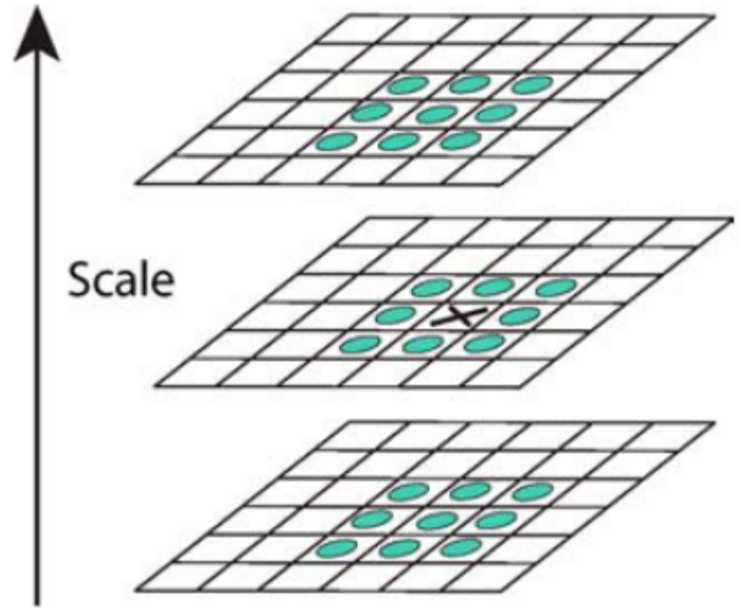
Building Gaussian Pyramid

- At each scale, σ is multiplied by k to produce multiple Gaussian levels.



Extrema Selection

- Each candidate point is compared with 26 neighbors (8 in current scale + 9 in scale above + 9 in scale below).
- Selected only if it is a local maximum or minimum → becomes a keypoint candidate.



Keypoint Localization

- Once a keypoint candidate has been found by comparing a pixel to its neighbors, the next step is to perform a detailed fit to the nearby data for location, scale, and ratio of principal curvatures. This information allows points to be rejected that have low contrast.
- This is done with 3 stages:
  1. Refining the Keypoint Location (Sub-pixel Accuracy)
  2. Eliminating Low-Contrast Points
  3. Eliminating Edge Responses

Keypoint Localization

1. Refining the Keypoint Location (Sub-pixel Accuracy)

- The initially found extrema are at integer pixel positions, but the true extremum may be between pixels.
- SIFT fits a **3D quadratic function** (Taylor expansion) to the local DoG values around the keypoint:

$$D(\mathbf{x}) = D + \frac{\delta D^T}{\delta \mathbf{x}} + \frac{1}{2}\mathbf{x}^T \frac{\delta^2 D}{\delta \mathbf{x}^2}\mathbf{x}$$

- where
  - $\mathbf{x} = (x, y, \sigma)$ is offsets vector from the sample point.
  - $\frac{\partial D}{\partial \mathbf{x}}$ is the gradient vector (a 3×1 matrix of first partial derivatives).
  - $\frac{\partial^2 D}{\partial \mathbf{x}^2}$ is the Hessian matrix (a 3×3 matrix of second partial derivatives)

Keypoint Localization

1. Refining the Keypoint Location (Sub-pixel Accuracy)

- Solving:
  - To find the true location of the extremum (where the derivative of the quadratic function equals zero), we take the derivative of $D(\mathbf{x})$ with respect to $\mathbf{x}$ and set it to zero:

$$\frac{\partial D}{\partial \mathbf{x}} + \left(\frac{\partial^2 D}{\partial \mathbf{x}^2}\right) \hat{\mathbf{x}} = 0$$

$$\hat{\mathbf{x}} = -\left(\frac{\partial^2 D}{\partial \mathbf{x}^2}\right)^{-1} \left(\frac{\partial D}{\partial \mathbf{x}}\right)$$

- gives the **sub-pixel** (and sub-scale) offset.
  - If $\hat{\mathbf{x}}$ is larger than 0.5 in any dimension, the keypoint is shifted to a nearby position and recomputed.

Keypoint Localization

2. Eliminating Low-Contrast Points

- If the value of the DoG function at the refined location is low: $\mid D(\hat{\mathbf{x}}) \mid < threshold\ (typically\ 0.03)$
- the keypoint is removed.

Keypoint Localization

3. Eliminating Edge Responses

- DoG produces strong responses along edges—these are not good stable keypoints.
- SIFT examines the **Hessian matrix** at the keypoint:

$$H = \begin{bmatrix} D_{xx} & D_{xy} \\ D_{xy} & D_{yy} \end{bmatrix}$$

- Eigenvalues of H (let's call them α and β, where α is the larger eigenvalue and β is the smaller one) tell whether the structure is:
  - blob-like (good keypoint) $\alpha \approx \beta$
  - edge-like (reject) $\alpha \gg \beta$

Keypoint Localization

3. Eliminating Edge Responses

- Instead of computing eigenvalues directly, SIFT uses the ratio:

$$\frac{(\mathrm{Tr}(H))^2}{\det(H)} < \frac{(r+1)^2}{r}$$

- $Tr(H)$ is trace of H, $\mathrm{Tr}(H) = D_{xx} + D_{yy} = \alpha + \beta$
- where r is the allowed edge ratio (typically 10).
- If this inequality fails, the keypoint lies on an edge → **discarded**.
- Hessian Matrix:
  - A **Hessian matrix** is a square matrix of **second-order partial derivatives** of a multivariable function.
  - It tells you how the function curves in different directions and is widely used in optimization, machine learning, and differential geometry.

Orientation Assignment

- To achieve rotation invariance, an orientation is assigned to each keypoint using gradient magnitudes and directions.

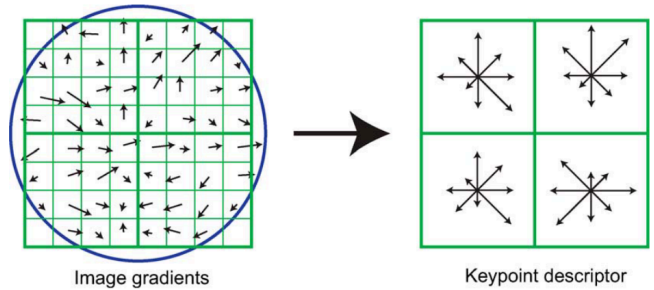
$$m(x,y) = \sqrt{(L(x+1,y) - L(x-1,y))^2 + (L(x,y+1) - L(x,y-1))^2}$$

- And the direction

$$\theta(x,y) = tan^{-1} \frac{L(x,y+1) - L(x,y-1)}{L(x+1,y) - L(x-1,y)}$$

Descriptor

- A descriptor is created by sampling gradients around the keypoint, weighted by a Gaussian window.
- These samples are grouped using a directional histogram on 4x4 sub-regions
- These are accumulated into 4×4 sub-regions with 8 orientation bins → 128-dimensional vector.



Descriptor Normalization

- The descriptor consists of a vector that contains the values of all entries in the orientation histogram, corresponding to the lengths of the arrows.
- The vector is normalized to reduce the effects of lighting changes by scaling it to unit length.
- Any change in image contrast that multiplies every pixel value by a constant will also multiply the gradients by the same constant, so this contrast change will be canceled out by normalizing the vectors.
- A change in brightness, where a constant is added to every pixel in the image, will not affect the gradient values, since they are computed from pixel differences.

Matching

- Descriptors are compared using Euclidean distance.
- For many points, nearest-neighbor matching is used.

SURF

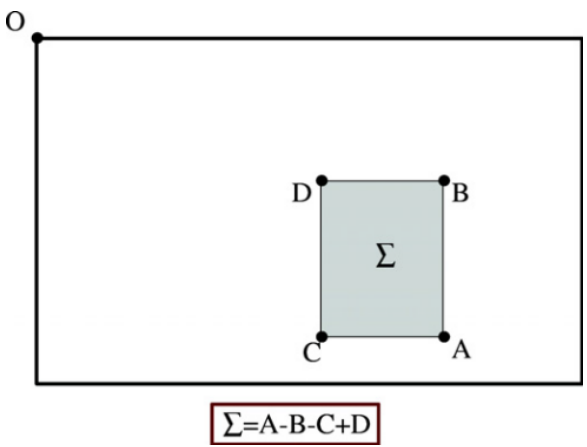
- Speeded Up Robust Features
- Faster alternative to SIFT.
- Uses integral images and Hessian determinant approximation for faster keypoint extraction.
- DoG (SIFT) → replaced by Hessian-based detector.

Integral Images

- Integral image: each pixel equals the sum of all previous pixels.

$$I_{\Sigma}(x, y) = \sum_{i \leq x, j \leq y} I(i, j)$$

- Converts summing over a rectangular region from O(n) to O(1):
  - Any rectangle sum can be computed with four corner values.



Keypoint Detection

- Uses Hessian matrix approximation:

$$H(x, \sigma) = \begin{bmatrix} L_{xx}(x, \sigma) & L_{xy}(x, \sigma) \\ L_{xy}(x, \sigma) & L_{yy}(x, \sigma) \end{bmatrix}$$

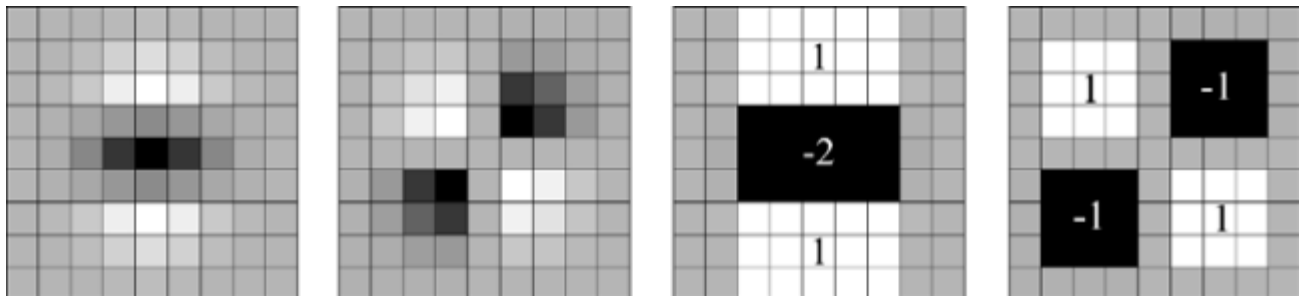
- Keypoints are detected using determinant:

$$\det(H) = L_{xx}L_{yy} - (L_{xy})^2$$

- Where $L_{xx}(x, \sigma)$ is the result of applying the second gaussian derivative $\frac{\delta^2}{\delta x^2} g(\sigma)$ and similarly for $L_{xy}(x, \sigma)$ and $L_{yy}(x, \sigma)$

Filter Approximation

- Second-order Gaussian derivatives are expensive for being based on floating point arithmetic.
- SURF approximates them with square-shaped filters to exploit integral images.



Approximate Determinant

- Approximate determinant:

$$\det(H_{approx}) = D_{xx}D_{yy} - (0.9D_{xy})^2$$

- σ controls filter size: 9×9, 15×15, 21×21, 27×27, etc. where the lowest value is $\sigma = 1.2$

Non-maximum Suppression

- 3×3×3 neighborhood suppression across scales selects strongest responses.
- Similar to SIFT scale-space maxima.

Descriptor Extraction

1. Orientation Assignment

- To achieve rotation invariance, we need to determine a direction we can use to reproduce the point, Haar wavelet responses in x and y directions are computed in a circular region of radius 6s.
- Responses are weighted using a Gaussian kernel.
- After computing the wavelet values at the keypoint center, they are normalized using a Gaussian weight of value $\sigma = 2.5$.
- The wavelet values are represented as vectors, where the horizontal value is projected onto the x-axis and the vertical value onto the y-axis.
- Gaussian normalization works slightly differently from traditional normalization:
  - In traditional normalization, all values are scaled by the same factor.
- In Gaussian normalization, the divisor changes and increases as we move further from the center of the image (the divisor follows a Gaussian relationship).

Descriptor Extraction

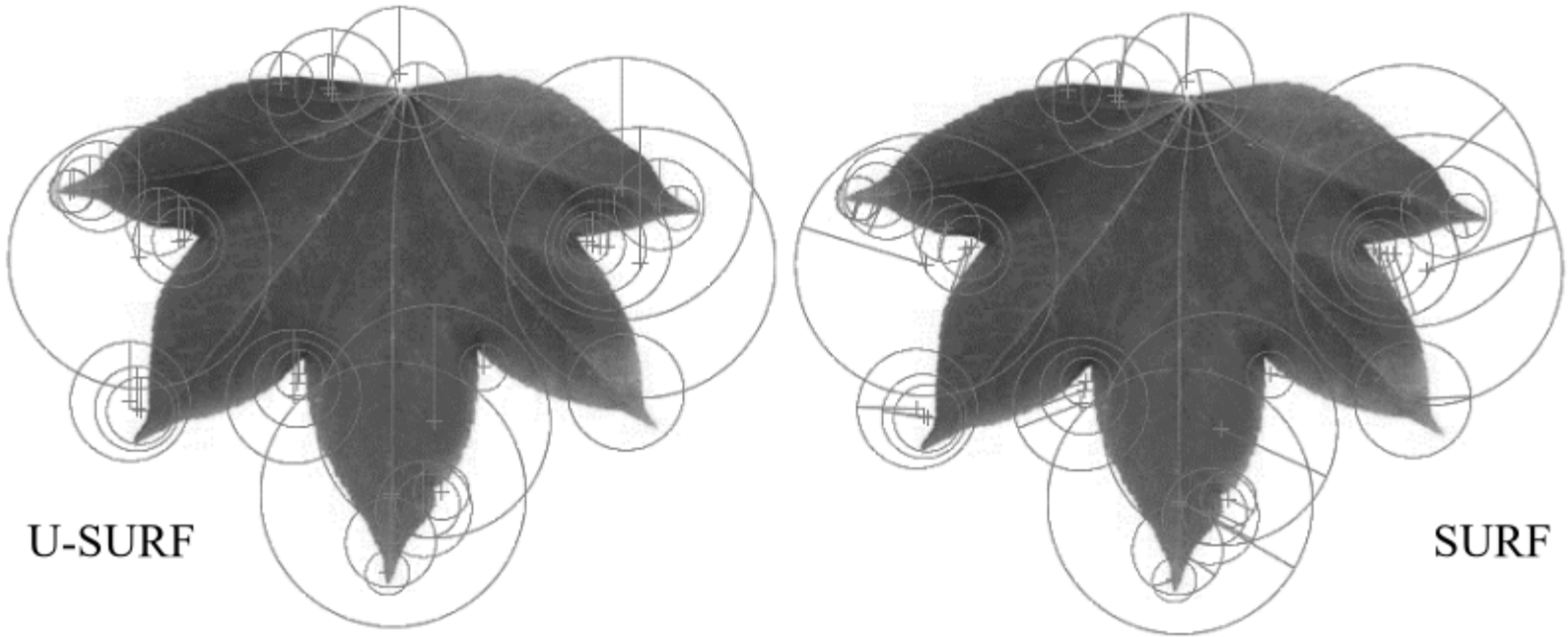
1. Orientation Assignment

- To compute the orientation, a sliding window of size $\frac{\pi}{3}$ is used, where the horizontal and vertical values within the window are summed.
- This produces a new vector, and the keypoint's orientation is taken as the direction of the longest vector.

Descriptor Extraction

Upright SURF (U-SURF)

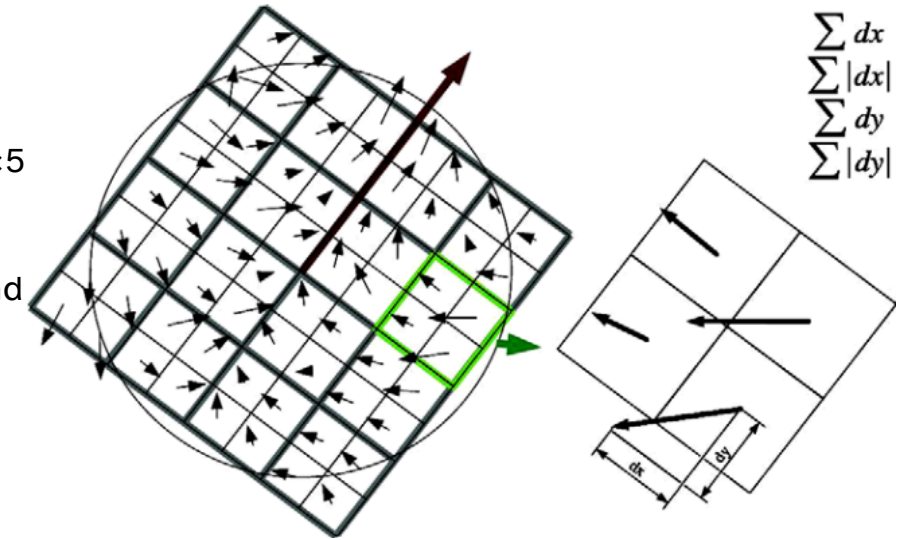
- A simplified form without orientation calculation.
- Faster and effective when camera rotation is limited.



Descriptor Extraction

2. Descriptor Construction

- A 20s×20s square region is built around the keypoint and rotated to match orientation.
- It is divided into 4×4 subregions each contains 5×5 samples.
- We calculate the sum of Haar wavelets on the X and Y axes as well as absolute sum on both axes
- $v = (\Sigma d_x, \Sigma d_y, \Sigma |d_x|, \Sigma |d_y|)$



Haar Wavelets

- Haar wavelets are square-shaped basis functions.
- They approximate both spatial and frequency representations.

Similarity Test

- Similarity is measured using Euclidean distance.
- Matching between two images uses nearest-neighbor matching.

